SIAI
2026/2027

Practical Assignment #1

Goals

DHCP configuration.

SSH.

DNS primary and slave server's configuration.

LDAP.

General description

In this assignment, we explore the DNS and DHCP services configuration to create two network infrastructure (*port* & *usa*) that will support the internal organisation communications and its services. Figure 1 illustrates the scenario for the practical assignment. For this assignment, you will need at least 4 +1 VMs, installed on your own machine (use a type 2 hypervisor such as VMware Workstation, UTM or Virtualbox). The two network´s infrastructure is formed by an internal sub-network and a DMZ sub-network. The DNS domain for this scenario is **dom.pt**, which will be used for the main services and two DNS sub-domains, the Port DNS domain **port.dom.pt**, and the Usa DNS domain **usa.dom.pt**. In this scenario, we also employ the DHCP service to automatically configure all network equipment as described next.

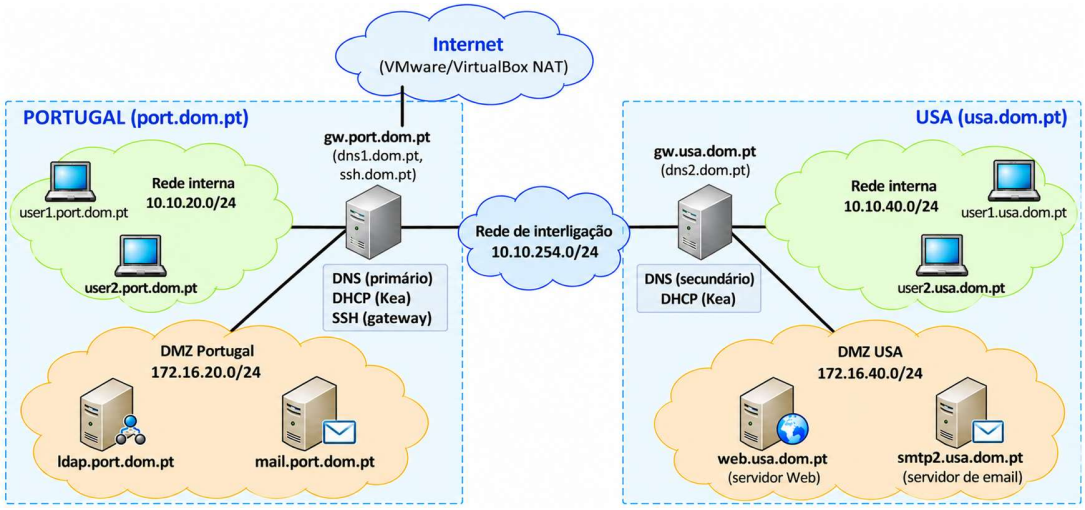


Figure 1: Assignment scenario

Network addressing and routing

The addressing plan is shown in Figure 1:

- PORT internal network: **10.10.20.0/24**; PORT DMZ: **172.16.20.0/24**.
- USA internal network: **10.10.40.0/24**; USA DMZ: **172.16.40.0/24**.
- Interconnection network between the two routers: **10.10.254.0/24**.
- gw.port.dom.pt is the only router connected to the Internet (hypervisor NAT) and performs NAT (masquerading) for all internal and DMZ networks, including the USA networks. gw.usa.dom.pt uses gw.port.dom.pt as its default gateway, and the static routes required for the four sub-networks to reach each other through the interconnection network.

DHCP server configuration

The DHCP service is required in the two network routers, gw.port.dom.pt and gw.usa.dom.pt, both using KEA DHCP and should be configured according to the following requirements:

- Clients in the PORT and USA networks (internal and DMZ) must obtain their network configuration via DHCP: IP address, subnet mask, default gateway, domain name and DNS servers.
- All servers should obtain the same IP address (except routers).
- The DHCP leases should be valid for 8 hours for dynamic DHCP clients and renewed after 6 hours (renew timer).
- Network names and DNS servers should consider the DNS configurations described next.
- Dynamic DNS: dynamic clients must be automatically registered in the DNS (forward and reverse zones) using the Kea DHCP-DDNS server (kea-dhcp-ddns). The dynamic updates sent to BIND must be authenticated with a TSIG key.

DNS service configuration

Our goal is to configure the DNS domain **dom.pt** and the sub-domains **port.dom.pt** and **usa.dom.pt**. These domains are supported by the two DNS servers, **dns1.dom.pt** and **dns2.dom.pt**, both using the BIND service. The server **dns1.dom.pt** will be the master and the server **dns2.dom.pt** will be the slave for the three DNS domains (3 zone configurations). Changes to the zones on the master must be immediately propagated to the respective DNS slave. The three DNS domains should be fully functional in both network´s infrastructure.

DNS domain **dom.pt**:

- This is the main DNS domain for this scenario and is intended to be responsible for the services available to the public, in this case, the DNS and mx´s services.
- The default domain server should resolve to the webserver, **web.usa.dom.pt**, which should be accessible through the following alternative names: **web.dom.pt** and **ftp.dom.pt**.
- Router PORT should be resolved through the following DNS names: **ssh.dom.pt** and **dns1.dom.pt**.
- Electronic mail sent to the **dom.pt** domain should be received by the hosts **mail.port.dom.pt** (preferred) and **smtp2.usa.dom.pt** (backup). The two sub-domains don't require email configuration.
- The mail server (**mail.port.dom.pt**) must be registered as **mail.dom.pt** and have the alternative names **webmail.dom.pt**, **pop.dom.pt** and **imap.dom.pt**.
- The LDAP server (ldap.port.dom.pt) must be reachable as **ldap.dom.pt**.
- Sign the **dom.pt** zone with DNSSEC (e.g. with the BIND `dnssec-policy` option) .

DNS domain **port.dom.pt**:

- Hosts in PORT internal and DMZ networks should be registered in the **port.dom.pt** domain, and the DNS servers authoritative for this domain are **dns1.port.dom.pt** (master) and **dns2.usa.dom.pt** (slave).
- Register all servers in the DNS: **gw.port.dom.pt**, **ldap.port.dom.pt**, **mail.port.dom.pt**.
- Configure the reverse DNS for the two PORT networks.

DNS domain **usa.dom.pt**:

- Hosts in USA internal and DMZ networks should be registered in the **usa.dom.pt** domain, and the DNS servers authoritative for this domain are **dns1.port.dom.pt** (master) and **dns2.usa.dom.pt** (slave).
- Register all servers in the DNS: **gw.usa.dom.pt**, **web.usa.dom.pt**, **smtp2.usa.dom.pt**.
- Configure the reverse DNS for the two USA networks.

LDAP configuration

For centralized user information, configure an LDAP server in **ldap.dom.pt** to manage the domain **dc=dom,dc=pt**. Configure your LDAP with the required schemas to address the assignment requirements. Create a group organizational unit **ou=group,dc=dom,dc=pt** with two users groups: managers, developers. Create a people organizational unit **ou=people,dc=dom,dc=pt** with four users, two managers and two developers. Each user must have at least, a uidNumber, gidNumber, home directory, login shell and password.

Configure the client machines in both **port.dom.pt** and **usa.dom.pt** to use the LDAP server for centralized user authentication. Each of the four LDAP users must be able to successfully log in to a client machine using their LDAP credentials. The authentication must be performed through the LDAP directory and not through an equivalent local user account. Protect the LDAP traffic with StartTLS or LDAPS, using a certificate issued by your own certificate authority; the clients must validate the server certificate.

SSH gateway

The PORT router (ssh.dom.pt) should be used as the SSH gateway to access the mail servers located in the PORT and USA networks. From this machine, it should be possible to access both mail.port.dom.pt and smtp2.usa.dom.pt using secure passwordless SSH authentication.

- On the mail servers, password authentication and direct root login must be disabled, and SSH connections are only accepted from ssh.dom.pt. From a client machine, the mail servers are reached through the gateway with a single command (`ssh -J`).
- Only members of the LDAP group **managers** may log in to the mail servers (e.g. with `AllowGroups`); developers must be refused. The mail servers must therefore also be LDAP clients.

Delivery of the Practical Assignment

The assignment MUST be developed and submitted by groups of 2 students.

With your assignment, you should deliver a report containing the following information:

- Descriptions of the configurations for the implementation of the assignment.
- A description of the tests performed to validate the various functionalities implemented.
- Remaining relevant information, including an explanation of the new DNSSEC record types (DNSKEY, RRSIG, NSEC) and of why the signed zone must be stored in `/var/lib/bind` instead of `/etc/bind`.

For the delivery of the assignment, put your report, as well as the relevant configuration files, in a single archive. Please note also the delivery deadline:

- The VMs used in the assignment are required for the assignment defence: they must be kept fully operational after the delivery, as the implemented functionalities will be demonstrated and tested live during the defence.
- The deadline for the assignment delivery (configuration files and report) is: 17/11/2026.
- Each day of delay will penalise its final score of 20%.

About the virtualized setup

- Use VMware Workstation for this assignment together with the Debian Linux deployments (the distribution used in our classes).